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22UOC205- STEM PROJECT

FIRE DETECTION

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ABSTRACT

This STEM project presents a smart fire detection system powered by an ESP32 development board. It features an MQ-2 gas sensor for smoke detection, an IR flame sensor for fire identification, and a DHT11/DHT22 sensor for temperature and humidity monitoring. A 16x2 LCD with I2C displays real-time status, while a passive buzzer and LED indicators provide local alerts. The setup is prototyped on a breadboard with jumper wires, powered via USB or external sources, making it cost-effective and adaptable for various applications.

PROBLEM STATEMENT

Traditional fire detection systems often suffer from delayed response, false alarms, and lack of real-time alerts or remote monitoring. There is a need for a smart, multi-sensor system that can accurately detect fires, alert users instantly, and enable automated emergency responses.

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LITERATURE SURVEY

1. Improved Fire Detection and Alarm Systems

Lawrence White, Raymond Ajax

Year: 2025

This paper introduces a smart fire detection and alarm system using machine learning, IoT, and multi-sensor fusion to enhance accuracy, reduce false alarms, and enable real-time monitoring for improved fire safety.

Drawbacks:

•Enhanced sensitivity might lead to increased false alarms, causing unnecessary evacuations and disruptions.

•Advanced detection technologies, such as AI-driven sensors, thermal imaging, or smart alarms, can be expensive to install and maintain.

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LITERATURE SURVEY

2. Fire Detection Method in Smart City Environments Using a Deep-Learning-Based Approach

Kuldoshboy Azavoz , Mukhriddin Mukhiddinov , Fazliddin Makhmudov ,Young Im Cho

Year: 2022

This study presents a vision-based fire detection system for smart cities using AI and deep learning. An improved YOLOv4 model enhances fire detection accuracy, recognizes small sparks within 8 seconds, and adapt to various weather conditions. The system offers real-time monitoring, ensuring rapid and reliable fire safety in urban areas.

Drawback :

- **High Computational Costs** – Deep learning models require significant processing power, which can be expensive and energy-intensive.

- **Data Quality & Availability** – The accuracy of deep-learning-based fire detection depends on large, high-quality datasets, which may not always be available.

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LITERATURE SURVEY

3. Fire Detection Using Multi Color Space and Background Modeling

A. Khalil, S.U. Rahman, F. Alam, I. Ahmad, I. Khalil
Year: 2021

This study presents a novel fire detection method using RGB and CIE color models, combining motion detection with fire object tracking to minimize false alarms and assess risk. By reducing parameters, the system improves accuracy while maintaining precision comparable to existing methods. Experimental results validate its effectiveness in fire detection.

Drawback :

• This method relies heavily on color-based detection, which can lead to **false positives** when objects or lighting conditions mimic the color of fire

•Background modeling can struggle in **dynamic environments**, especially outdoors where conditions change due to weather, wind, or illumination shifts.

LITERATURE SURVEY

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4.Active Fire Detection for Fire Emergency Management

J. San-Miguel-Ayanz, N. Raval
Year: 2005

Active fire detection uses thermal sensors on satellites, aircraft, and fire-watch towers to locate fires by detecting hot spots or smoke plumes. While effective, limitations include low revisit frequency and sensor accuracy in forest fire-fighting. Future multi-satellite systems aim to improve real-time monitoring

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Drawback:

Setting up and maintaining active fire detection systems can be **expensive**, especiall for large-scale buildings or industrial sites.

Sensors may trigger **false alarms** due to environmental factors like steam, dust, or sudden temperature changes

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EXISTING SYSTEM

- Sensors: MQ-2 smoke sensor, DHT11 temperature sensor, and IR flame sensor.
- Microcontroller: Arduino
- Simple threshold-based logic to detect fire (e.g., if temperature > 60°C or smoke level > 500 ppm, trigger alert).
- Cloud Platform: ThingSpeak for data storage and visualization.

EXISTING SYSTEM

Drawbacks:

- Prone to false alarms as it relies solely on threshold values without cross- verification.
- Limited scalability due to dependency on stable Wi-Fi connectivity.
- High power consumption, making it unsuitable for remote or offgrid areas.
- Lack of advanced analytics to differentiate between fire and other heat/smoke sources.

These existing systems highlight the progress in fire detection technology but also underscore the need for a more balanced solution that addresses cost, scalability, false positives, and connectivity challenges for broader applicability

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PROPOSED SYSTEM

- This smart fire detection system is built around an ESP32 development board, which processes data from multiple sensors to ensure accurate hazard detection. It incorporates an MQ-2 gas sensor for smoke and gas detection, an IR flame sensor module for fire identification, and a DHT11/DHT22 sensor to monitor ambient temperature and humidity. A 16x2 LCD display with an I2C module presents real-time system status, while a passive buzzer and LED indicators provide immediate local alerts.

- The setup is prototyped on a breadboard, utilizing jumper wires for flexible connectivity. Power is supplied via a USB cable, with support for external 5V DC sources, battery backup, and an optional soar panel to enhance reliability for off-grid applications. The system ensures robust performance through intelligent data processing and threshold verification, improving detection accuracy while minimizing false alarms.

- Designed for efficiency and adaptability, this hardware-based fire detection system is ideal for homes, schools, and small businesses, providing an effective solution for fire safety.

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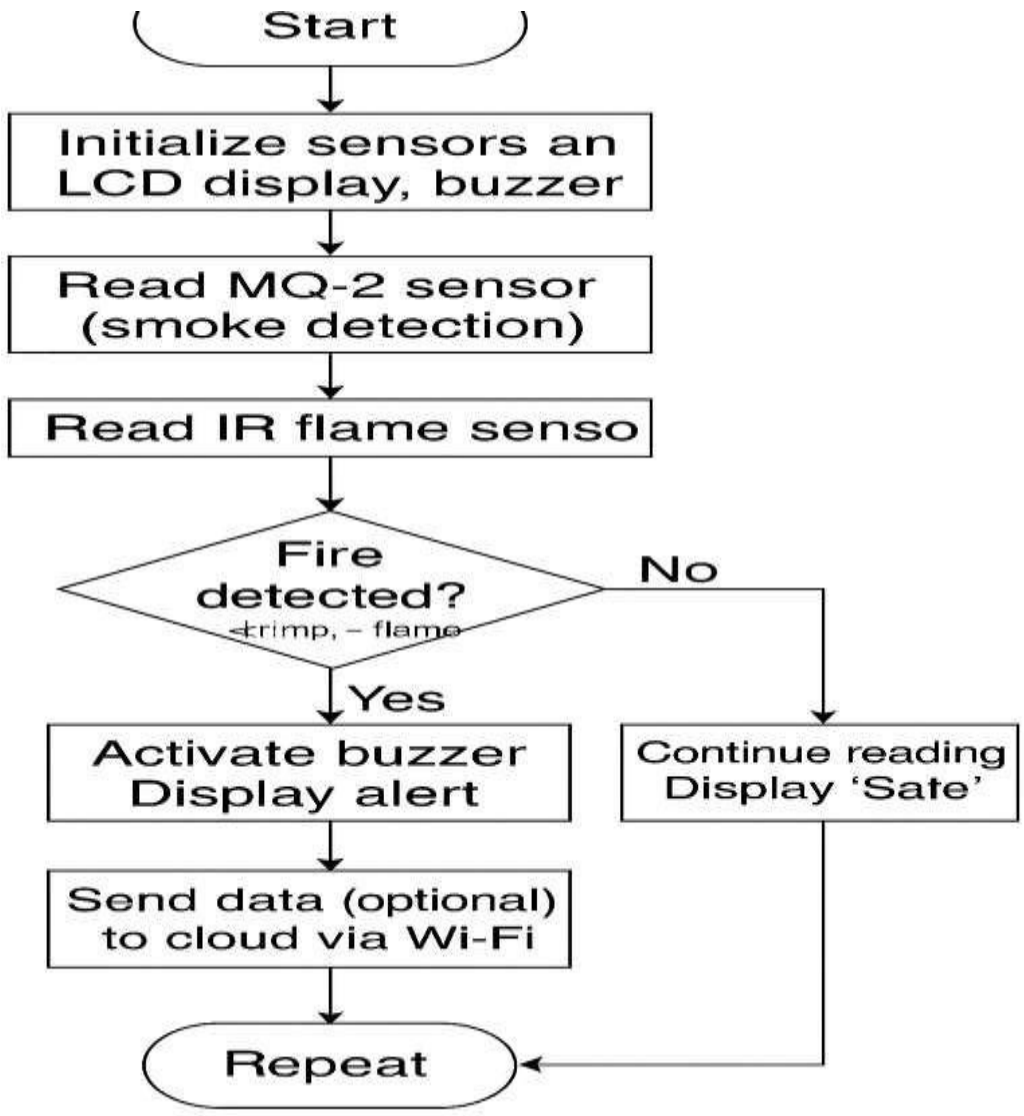
PROPOSED SYSTEM ARCHITECTURE



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List of Hardware

- Sensors: MQ-2 (smoke), DHT22 (temp/humidity), IR flame sensor.
- Microcontroller: ESP32 (processing, Wi-Fi).
- Breadboard
- 16x2 LCD Display
- Jumper wire
- Passive Buzzer
- USB cable
- Power: 5V DC

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LIST OF MODULES

Module 1:
(Sensor Setup)

- DHT11/DHT22 for temperature & humidity
- MQ2 for gas detection
- IR flame sensor for fire detection

Module 2:
(Microcontroller Integration & visual Output)

- Connect sensors to ESP32
- Use LEDs for alerts
- Display readings on 16×2 LCD

Module 3:
(Circuit Assembly)

- Use jumper wires for connections
- Power up the system
- Test the complete setup

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MODULE 1 DESCRIPTION

In this module, the primary focus is on setting up the environmental sensors required for fire detection. The MQ2 gas sensor is integrated to detect the presence of combustible gases such as LPG, methane, or smoke. Additionally, an IR flame sensor is configured to detect the presence of flames or fire in the environment. These sensors work together to monitor critical conditions that may indicate a fire hazard.

MODULE 2 DESCRIPTION

In this module, all sensors are connected to the ESP32 microcontroller, LEDs are used to provide alert signals, and the sensor data is displayed on a 16×2 LCD screen.

MODULE 3 DESCRIPTION

In the final module, the complete hardware setup is assembled on a breadboard using jumper wires to establish all necessary electrical connections between components. The system is powered using a 5V DC supply or USB connection to the ESP32. Once the setup is complete, the circuit is tested thoroughly to ensure that all sensors are working correctly

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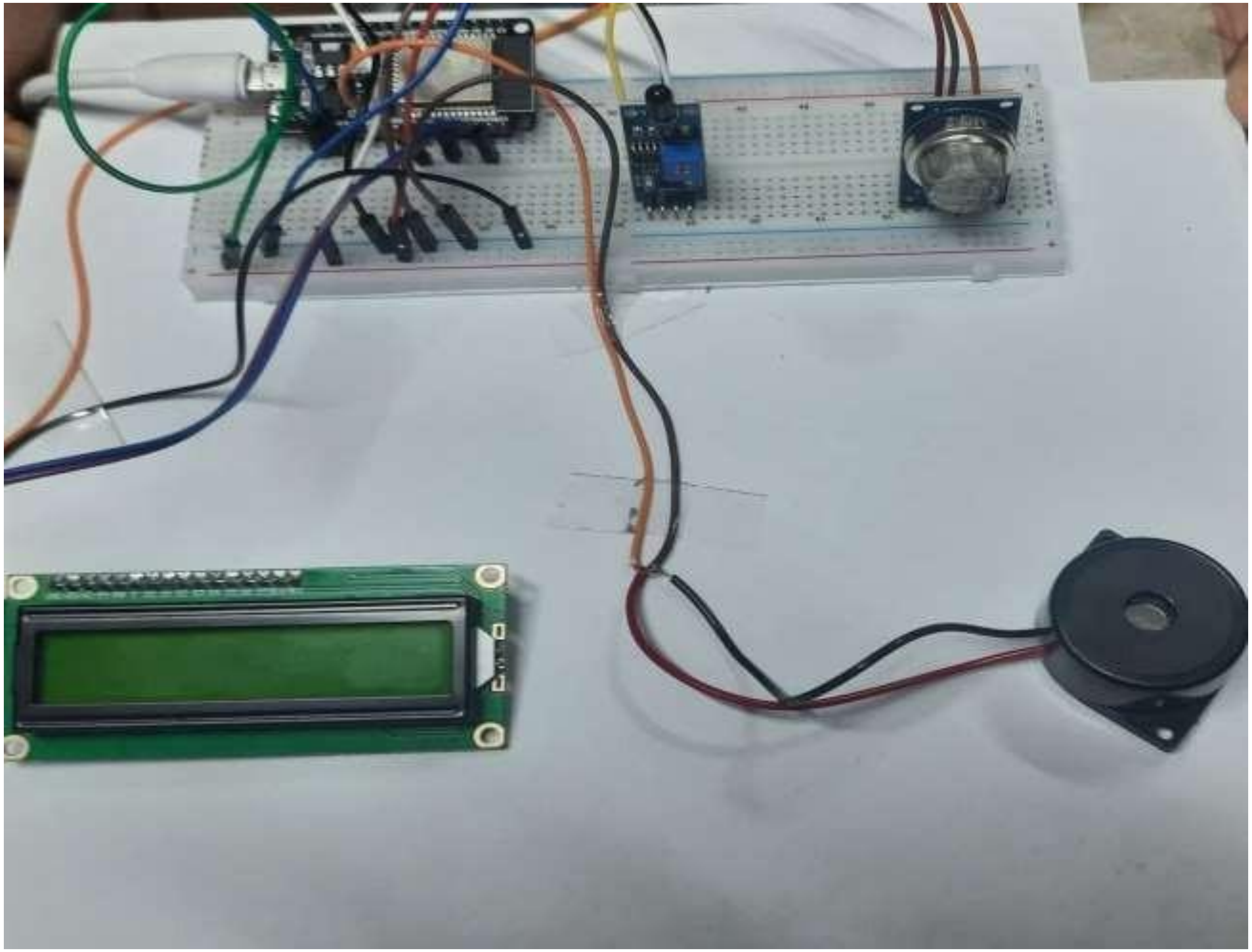
MODULE 1 OUTPUT SCREENSHOT



CONCLUSION

In this project, we successfully designed and developed a fire detection system using sensors and microcontrollers to identify early signs of fire through heat and smoke. The system demonstrated quick response times and effectively triggered alarms when fire indicators were detected. This project highlights the importance of combining technology and engineering to create practical safety solutions. Our results show that such a system can be cost-effective and scalable for use in homes, schools, and workplaces, contributing to improved fire safety and prevention.

REFERENCE

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Thank you